

VANI 3-Node Integration — Process Overview

How one radio intercept flows through NODE-A (denoise + diarization), NODE-B (language + dialect ID), and NODE-C (VANI orchestrator). Generated 2026-07-09.

1. The three machines

Node	Owner	Role	Address
NODE-A	Gaurav	Denoise (DeepFilterNet3) + diarization (DiariZen)	192.168.10.11:8801
NODE-B	Sanket	Language ID (8) + Mandarin dialect (MMS-LID-4017)	192.168.10.12:8802
NODE-C	this repo	VAD, ASR, translation, ISUM, GUI, orchestrator	192.168.10.13:8501

- Isolated /24 LAN, one switch, no internet / gateway / DNS.
- A and B are **stateless** HTTP services; they know nothing of each other or of VANI.
- **NODE-C is the only orchestrator** — it calls A and B and assembles the result.

2. End-to-end flow for one clip

```

Analyze uploads clip.wav in the Streamlit GUI [NODE-C]
|
1. VAD - strip silence, keep speech regions (Silero) [NODE-C]
|
| -> vad.wav
2. Coarse MMS-LID probe on vad.wav [NODE-C]
|
| -> coarse language hint (e.g. "zh") -> Gaurav tag "mandarin"
3. POST /process?lang=mandarin&variant=robust (raw WAV) -----> [NODE-A]
|
| <--- zip: diarization.json, summary.json, speaker + diarizes
|
| mixed_denoised.wav, Speaker_N_Denoised.wav
4. Normalize A's output; use mixed_denoised.wav as the audio [NODE-C]
|
| (skip C's local denoise/bandpass; keep normalize)
5. POST /api/analyze (each Speaker_N_Denoised.wav) -----> [NODE-B]
|
| <--- per speaker: {top1_language, confidence, dialect} B: LID+dialect
6. Aggregate B's per-speaker LID -> dominant language by talk time [NODE-C]
|
| -> picks the language-specific Whisper model
7. Chunk mixed_denoised.wav (VAD-aware) [NODE-C]
8. ASR - Whisper large-v3 CT2 (or SeamlessM4T for pa/ne), [NODE-C]
|
| language forced from B's answer -> transcript + segments
9. Label each ASR segment by time-overlap vs A's diarization.json [NODE-C]
|
| -> SPEAKER_A / SPEAKER_B / ...
10. LanguageRouter vote: B's LID = audio vote, FastText = text vote [NODE-C]
11. Translation - NLLB-200 (IndicTrans2 for Dogri) -> English [NODE-C]
12. Keyword detection -> threat level + categories [NODE-C]
13. ISUM - Ollama gemma3:4b -> structured intelligence summary (5W) [NODE-C]
14. Persist to SQLite, render in GUI [NODE-C]

```

3. Why the ordering is what it is

A circular dependency between the partners, resolved for free by NODE-C's existing probe:

- **NODE-A needs a language tag before it runs** (per-language clustering; "default" measurably hurts). It does no language ID itself.
- **NODE-B wants to run after NODE-A** (trained on a post-denoiser, non-reverberant distribution — it expects clean per-speaker tracks).

Resolution: NODE-C runs its local MMS-LID probe early (on vad.wav) to give NODE-A the coarse tag; NODE-B then runs on NODE-A's clean per-speaker tracks for the authoritative answer. Neither partner changes to accommodate the other.

4. What each node does internally

NODE-A — who spoke when, and clean it up. Loads DiariZen + DeepFilterNet3 once at startup (a ~90 s cold call becomes ~5–10 s warm); diarizes; denoises each speaker track independently (preserves speaker identity, 15.62% EER); reconstructs **mixed_denoised.wav** on the original wall-clock timeline; serializes GPU work behind a lock (returns 503 when busy). The clean/robust variant is not interchangeable.

NODE-B — what language, which Mandarin dialect. One MMS-LID-4017 forward pass → 8-way language + 7-way dialect distribution. Trained without reverb (runs after A). Analyzes the first 10 s only; NODE-C windows longer tracks. RMS<0.01 silence gate. Frozen v1 language + v1 dialect pair (macro-F1 0.9764 / 0.601).

NODE-C — everything else + orchestration. VAD, coarse probe, chunking, ASR, diarization labeling, language vote, translation, keywords, ISUM, storage, GUI. Normalizes A's audio before forwarding to B (closes the gain-convention mismatch).

5. The two handoffs

```

NODE-C -> NODE-A
POST http://192.168.10.11:8801/process?lang=<mandarin|urdu|punjabi|pashto|default>
      &variant=<clean|robust>&mode=diarization-guided
      Content-Type: audio/wav      body = raw 16 kHz mono WAV bytes
200 -> application/zip { diarization.json, summary.json,
      mixed_denoised.wav, Speaker_1_Denoised.wav ... }
503 -> busy (GPU lock held)

NODE-C -> NODE-B
POST http://192.168.10.12:8802/api/analyze multipart file = one Speaker_N_Denoised.wav
200 -> { top1_language, top1_language_confidence,
      top1_dialect, top1_dialect_confidence, dialect_engaged, language_probs }
200 -> { no_speech_detected: true, rms }      (silence gate)

```

Both directions stay 16 kHz mono end to end, so nothing is ever resampled twice.

6. Language-code mapping

VANI → NODE-A (Gaurav's clustering knob):

VANI	Gaurav tag
zh	mandarin
ur	urdu
pa	punjabi
ps	pashto
anything else	default (logged — no tuned operating point)

NODE-B → VANI (Sanket's 8 classes):

Sanket	VANI	Fine-tuned VANI ASR?
urdu	ur	yes
pashto	ps	yes
kashmiri	ks	yes
dogri	doi	no — default Whisper + IndicTrans2
punjabi	pa	yes (routed to SeamlessM4T)
mandarin	zh	yes
cantonese	zh	no yue model -> zh
tibetan	bo	no — default Whisper

Critical asymmetry: hi (Hindi) and ne (Nepali) are not in NODE-B's 8-class set, but VANI supports both. NODE-B's answer is accepted only when confident AND mappable; otherwise NODE-C defers to its local MMS-LID probe. The local probe is never removed — it is the safety net for Hindi, Nepali, and for NODE-B being down.

7. The safety net — why this cannot regress the demo

- **NODE-A unreachable** → NODE-C runs local denoise + MFCC diarization.

- **NODE-B unreachable / unsure** → local MMS-LID vote carries the decision.
- **Both down / Standalone** → NODE-C is the original single-machine VANI, byte-for-byte.

Network mode (one build, either environment) — an Auto / Standalone / Networked switch backed by a once-per-session health probe:

Mode	Behaviour
Auto (default)	Probe A and B once; use whichever are reachable; else run fully local. Unreachable nodes skipped (no per-file latency).
Standalone	Never touch the network — pure local pipeline.
Networked	Trust the configured LAN nodes; fall back per-node on failure.

8. Roles at a glance

Concern	NODE-A	NODE-B	NODE-C
Denoising	per-speaker	-	fallback (local)
Diarization	yes	-	fallback + labeling
Language ID	needs a tag	authoritative	coarse probe + vote + fallback
Dialect (Mandarin)	-	yes	display
VAD / chunking	-	-	yes
ASR / translation	-	-	yes
Keywords / ISUM	-	-	yes
Orchestration / GUI / storage	-	-	yes